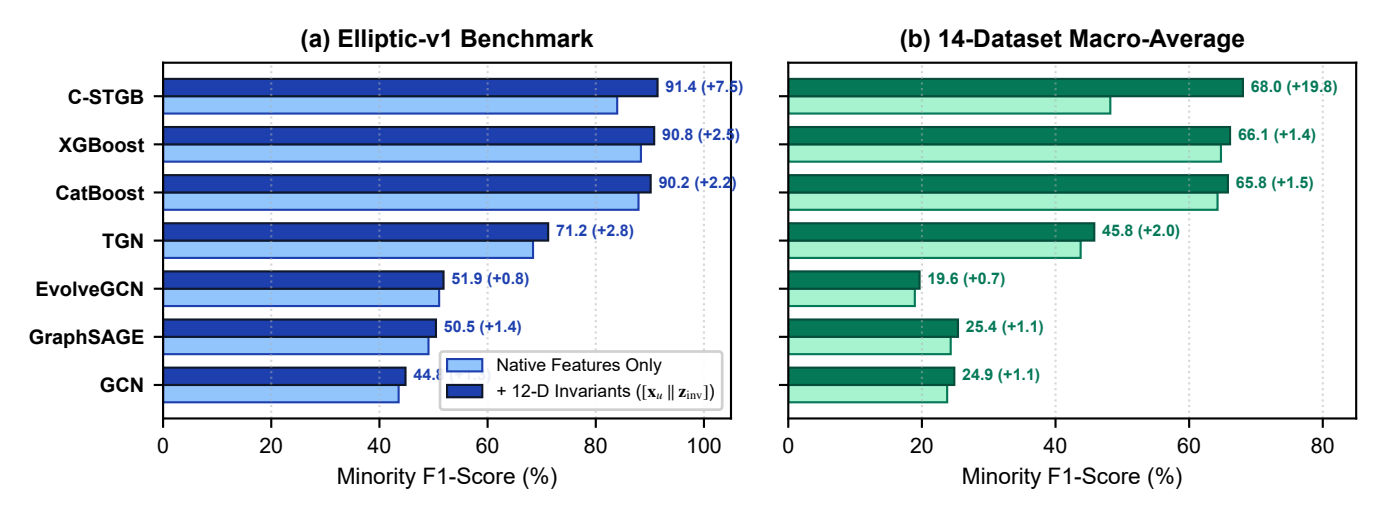

(a) Elliptic-v1 Benchmark
C-STGB
XGBoost
CatBoost
TGN
EvolveGCN
GraphSAGE
GCN
91.4 (+7.5)
90.8 (+2.8)
90.2 (+2.2)
71.2 (+2.8)
51.9 (+0.8)
50.5 (+1.4)
Native Features Only
+ 12-D Invariants ([$\mathbf{x}_u \,\|\, \mathbf{z}_{\text{inv}}$])
0
20
40
60
80
100
Minority F1-Score (%)
(b) 14-Dataset Macro-Average
68.0 (+19.8)
66.1 (+1.4)
65.8 (+1.5)
45.8 (+2.0)
19.6 (+0.7)
25.4 (+1.1)
24.9 (+1.1)
0
20
40
60
80
Minority F1-Score (%)